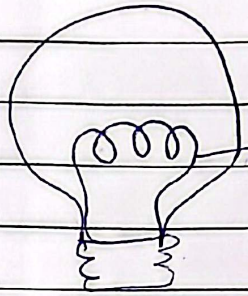
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Robotics:-

yt Video # 2 (Paul Mc Worter)

How light bulb works:-



→ Here electricity comes, & 95% converted to heat & only 5% converted to light

so its alot of wastage of energy, To solve this we use LEDs

To understand this, we first need to understand semiconductors

To understand semiconductors, lets talk about conductors

Semiconductor → You can manufacture it and customise it to be a conductor or insulator depending on your electric signal

Most electronic devices are built in semiconductor materials that

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uses Silicon.

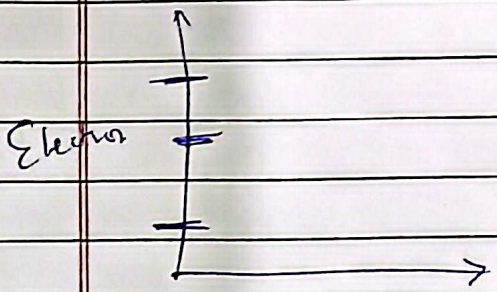
↳ ex: computer chips
microcontrollers

So almost everything in world of computers is based on semiconductors

So semi

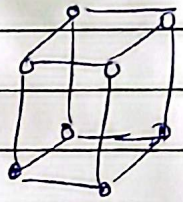
In atoms:

↳ electrons are at a specific energy level (states)



And semiconductors are actually crystals

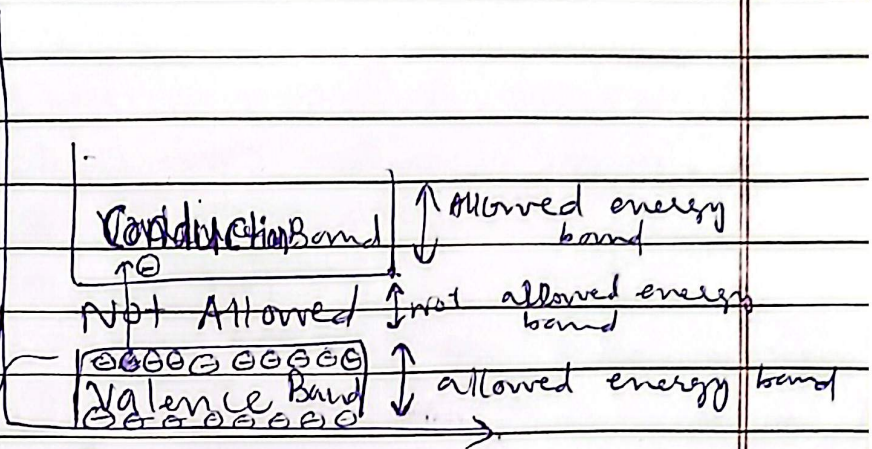
& these atoms are in perfect symmetry



And as we brings atoms together to form a crystal. & when atoms

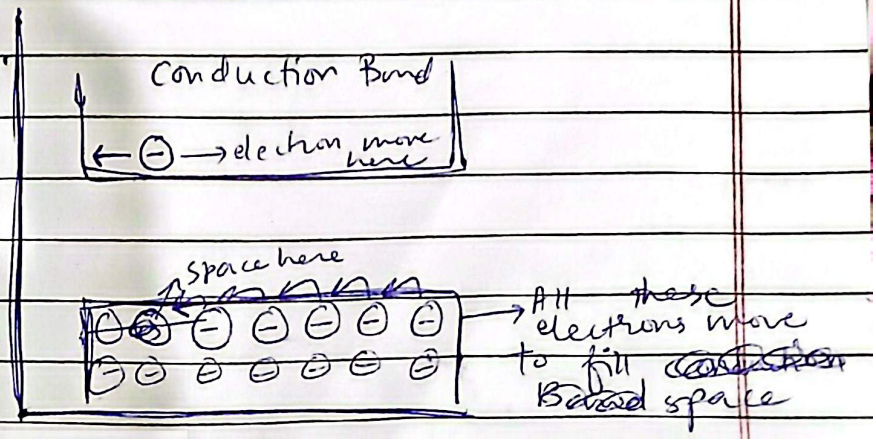
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Smears, the electrons instead of discrete energy levels, comes in allowed bands



So first electrons are in valence Band

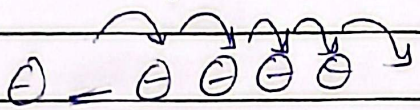
When we apply a certain thermal energy, then one lucky electron goes from Valence Band to conduction Band, Now when ~~there~~ electron in conduction Band move & we get conduction & also electrons in Valence Band come to proper positions in valence Band. so electrons also move there & hence conduction



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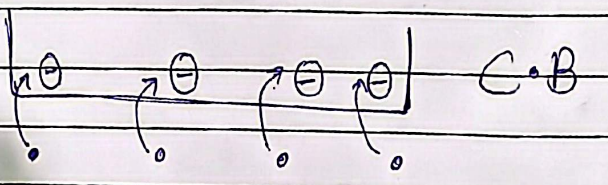
→ it easier to move one hole, instead of millions of electrons

But hole can also move instead of electrons moving



holes ⟶ moves upward (uphill)
electrons ⟶ moves downward (downhill)

Q# What if we place a very specially designed impurity atoms in this crystal?



called n-type



impurity atoms,

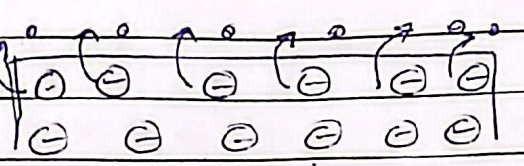
and they have just little energy diff with conduction Band & it is very easy for them to donate electrons, hence they are donor Atoms.

Semiconductor acts like a conductor.

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⇒ P-type



Now this time, the impurities are near the valence Band & the impurity atoms, so have a vacancy of electrons this time so they are gonna accept electrons. from valence Band

↳ holes are left behind in Valence Band.

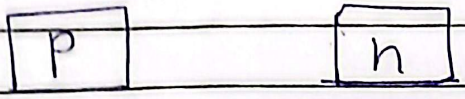
Intrinsic Semiconductors:

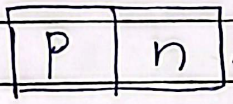
↳ No impurities, valence Band is full, conduction Band is empty, no current flow except just the Random ones, luckily generated.

↳ Now how all of this leads to (LED)

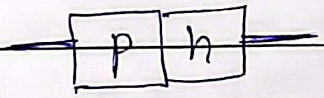
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Q # What if I bring a P-type material next to an n-type material?



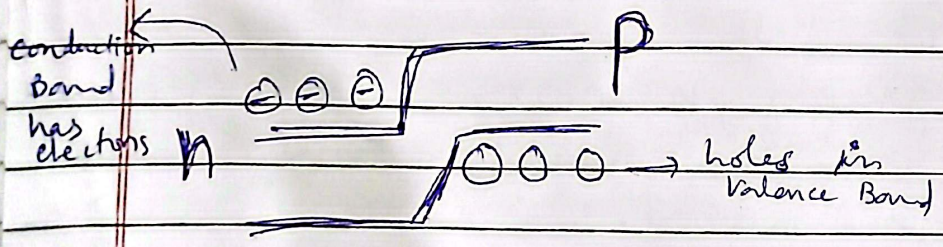


→ manufactured in a way that donor atoms at one side & acceptor atoms at other side



(*) Now lets look at energy bands & what energy bands are gonna look likes

→ you'd get an off set in the Band



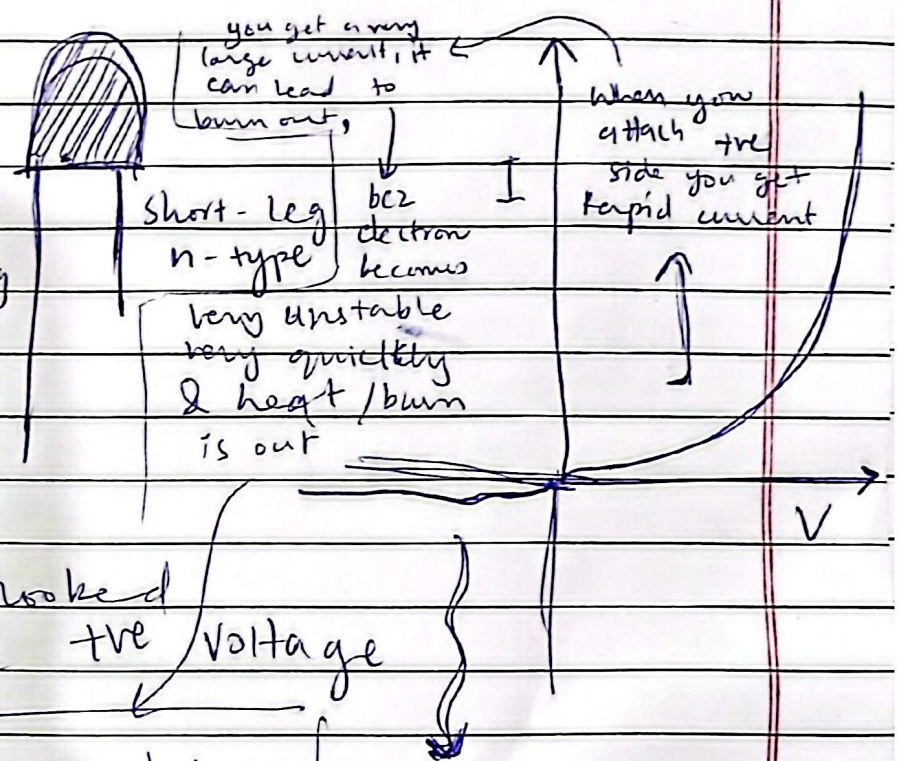
(*) now lets apply a voltage on P side (+ side of voltage) (− side) on n side

Date:

Now when we attach a negative side of voltage to the np band it increases the gap but attaching the +ve positive lowers the gap

So diode only works in one direction

↳ so hook it up correctly & put it in circuit correctly



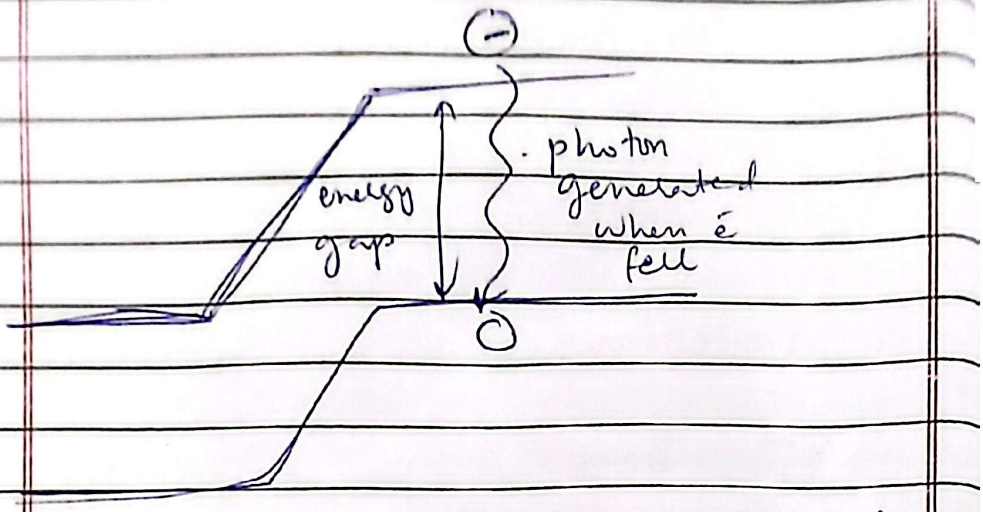
always hooked towards +ve voltage

To prevent burn, you always attach a limiting Resistor with diode

But when you attach a -ve voltage side, you get almost no current, just a little leakage of current

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Colors of LEDs



E_{photon} = E = related to frequency = hf

→ The larger the gap b/w the bands, the higher the frequency generated by photon

$$\rightarrow f = \frac{1}{\lambda}$$

frequency is inversely related to wavelength.

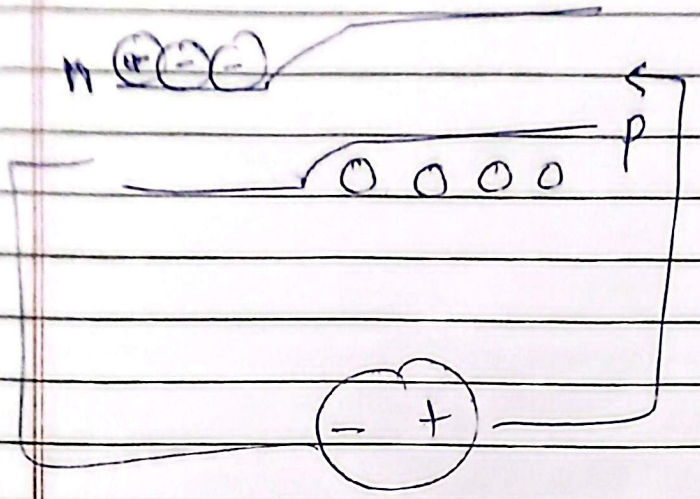
The higher the frequency, the lower λ is gonna be

→ ex: $f \uparrow \Rightarrow \lambda \downarrow \Rightarrow$ purple, violet, U.V

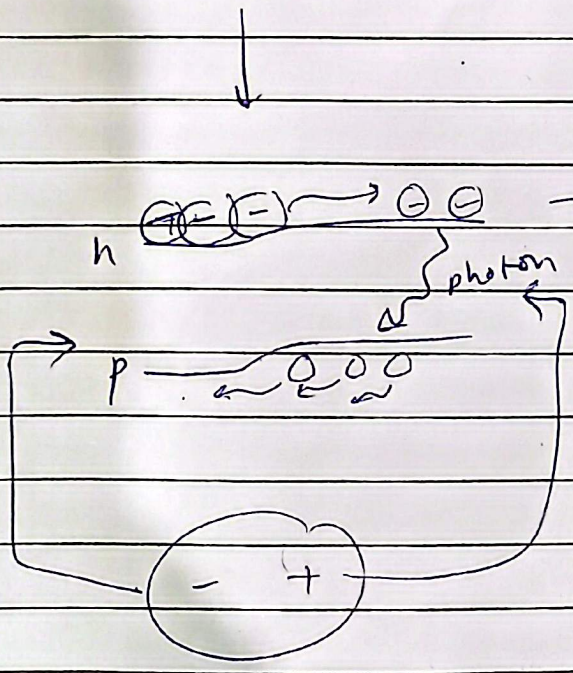
$f \downarrow \Rightarrow \lambda \uparrow \Rightarrow$ red, infrared

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(1) How connecting +ve voltage, makes the LEDs (diode) glow?



so the side makes the barrier low & it is easier for e⁻s & holes to move



→ now when electrons move they have extra energy, so it tries to transfer energy to holes (lower side) & photon is emitted, that photon generates light